

Tissue-Specific Radiation Responses

Key Results & Future Analysis Directions

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Central Question

Do different Arabidopsis tissues respond differently to ionizing radiation? This analysis used 6 NASA GeneLab radiation experiments (158 samples, 3 radiation types, 4 dose levels, 3 genotypes) and the AtGenExpress developmental atlas to assign differentially expressed genes (DEGs) to their predominant tissue. The answer is yes - tissue-specific responses are statistically significant in 4 of 6 comparisons, and the response profile differs across radiation conditions.

1. The Five Biggest Results

Result 1: Tissue enrichment is statistically significant in 4 of 6 comparisons

The chi-square test of independence (organ x DEG status) was significant in 4 comparisons with sufficient DEGs, confirming that radiation does not affect all tissues equally:

Comparison	Chi-sq	p-value	Enriched organs (DEG rate)
radiation_effect (Co-60 100Gy)	233.6	2.2e-49	Seedling 12.7%, Leaf_Shoot 8.1%
genotype_interaction (sog1-1)	14.4	0.006	Flower 3.1% enriched, Leaf_Shoot depleted
Cs137_100cGy	53.9	5.7e-11	Root 2.4%, Seedling 3.0% enriched
Cs137_10cGy	44.4	5.3e-9	Root 0.9%, Seedling 1.6% enriched
GCR40 (2 DEGs)	SKIP	-	Too few DEGs for testing
GCR80 (11 DEGs)	SKIP	-	Too few DEGs for testing

Interpretation

The strongest tissue bias is in the Co-60 100 Gy comparison, where Seedling genes are 3.7x more likely to be DEGs than the genome average (OR=3.67, padj=1.5e-30). This makes biological sense: the experiments used whole seedlings, so seedling-expressed genes are the most directly affected. Root and Seedling are consistently enriched across Cs-137 comparisons, while Flower is consistently depleted.

Result 2: sog1-1 reverses the direction of the radiation response

The genotype x radiation interaction comparison (OSD508+510) tests whether the sog1-1 DNA-damage-response mutant changes how plants respond to radiation. The result is striking: while the radiation main effect is 93% upregulated, the interaction DEGs are 82% downregulated.

The chi-square test for direction confirms this is highly significant: $X^2=456.5$, $df=1$, $p=2.8e-101$. This is the strongest statistical signal in the entire analysis.

Biological Meaning

SOG1 is the master transcription factor for DNA damage response in plants. When it is mutated (sog1-1), the plant loses its ability to upregulate radiation-responsive genes. The 433 interaction DEGs represent genes whose radiation response depends on SOG1 - they are upregulated in WT but not in the mutant. The tissue distribution also shifts: Flower and Seed_Silique show the most interaction DEGs, while Seedling (which dominated the radiation main effect) drops to third place.

Result 3: Clear dose-response for Cs-137 gamma radiation

The Cochran-Mantel-Haenszel test (stratified by organ) shows that 100 cGy produces 2.8x more tissue-specific DEGs than 10 cGy (common OR=2.846, $p=5.6e-13$). The enriched tissues (Root and Seedling) are the same at both doses - only the magnitude changes, not the tissue identity.

Why CMH instead of a simple chi-square?

The CMH test stratifies by organ, meaning it first compares 10cGy vs 100cGy within each organ separately, then combines the results. This removes the confounding effect of organs having different baseline DEG rates. A simple pooled chi-square would conflate tissue differences with dose differences.

Result 4: Radiation type matters - Co-60 vs Cs-137 produce different tissue responses

The CMH test comparing Co-60 100 Gy to Cs-137 100 cGy (stratified by organ) gives common $OR=0.328$ ($p=1.1e-39$), meaning Co-60 produces 3.1x more tissue-specific DEGs. However, this comparison is confounded by dose: 100 Gy = 10,000 cGy, which is 100x higher than the Cs-137 100 cGy dose. The tissue enrichment pattern also differs: Co-60 enriches Seedling and Leaf_Shoot, while Cs-137 enriches Root and Seedling.

The overall chi-square test confirms that tissue response profiles differ significantly across all 4 testable comparisons ($X^2=209.2$, $df=12$, $p=4.0e-38$).

Result 5: Simulated GCR produces almost no 2-fold transcriptional changes

The simulated Galactic Cosmic Radiation comparisons (OSD658) yielded only 2 DEGs at 40 cGy and 11 DEGs at 80 cGy - far too few for tissue enrichment statistics. This is a real biological signal, not a data quality issue. Mixed-particle beam radiation at these low doses produces minimal transcriptional changes that meet the 2-fold threshold.

Caveat

The near-zero DEG count for GCR may partly reflect the strict $|\log_2FC| \geq 1$ threshold. Lowering to $|\log_2FC| \geq 0.5$ would capture more genes but at the cost of biological interpretability. An alternative approach would be gene set enrichment analysis (GSEA), which does not require a hard threshold and can detect coordinated small changes across pathways.

2. Summary of All Statistical Tests

Per-comparison tests (4-test battery per comparison):

Comparison	Test	Stat	p-value	Key result
radiation_effect	Chi-sq independence	233.6	2.2e-49	Yes
radiation_effect	Fisher: Seedling enriched	-	1.5e-30	OR=3.67
radiation_effect	Fisher: Root depleted	-	5.8e-7	OR=0.56
radiation_effect	Direction chi-sq	62.7	8.0e-13	Root 97% up
genotype_interaction	Chi-sq independence	14.4	0.006	Yes
genotype_interaction	Direction chi-sq	47.7	1.1e-9	Root 89% down
Cs137_100cGy	Chi-sq independence	53.9	5.7e-11	Yes
Cs137_100cGy	Fisher: Root enriched	-	9.3e-4	OR=1.82
Cs137_10cGy	Chi-sq independence	44.4	5.3e-9	Yes
Cs137_10cGy	Fisher: Seedling enriched	-	7.5e-5	OR=3.74
GCR40	All tests	SKIP	-	2 DEGs only
GCR80	All tests	SKIP	-	11 DEGs only

Cross-comparison tests:

Test	Stat	p-value	Interpretation
CMH: Co-60 vs Cs-137 (by organ)	173.7	1.1e-39	OR=0.328 (Co-60 3.1x more)
CMH: Cs-137 10 vs 100 cGy (by organ)	52.0	5.6e-13	OR=2.846 (dose-response)
Chi-sq: genotype organ distribution	87.1	5.5e-18	Different tissue profiles
Chi-sq: direction (rad vs interaction)	456.5	2.8e-101	Up vs down dominant
Chi-sq: overall (4 comparisons)	209.2	4.0e-38	Profiles differ

3. Future Analysis Directions

The current analysis answers the question 'are there tissue-specific responses?' with a clear yes. The following directions would deepen understanding of how and why tissues differ.

Direction 1: Gene Ontology and pathway enrichment per tissue

Currently, we know which tissues are enriched for DEGs, but not what biological processes those DEGs are involved in. Running GO/KEGG enrichment separately for each organ's DEG set would reveal whether different tissues activate different pathways. For example:

- Root DEGs may be enriched for DNA repair genes (Root is enriched in Cs-137)
- Seedling DEGs may be enriched for stress response genes
- Flower interaction DEGs may reveal SOG1-dependent reproductive tissue pathways

Tools: clusterProfiler (R), agriGO (web), or DAVID. Input: the per-organ DEG CSVs already exported in `tissue_specific_degs_v2/`.

Direction 2: Gene Set Enrichment Analysis (GSEA) for low-DEG comparisons

GCR40 and GCR80 had too few DEGs (2 and 11) for tissue enrichment tests. GSEA does not require a hard DEG threshold - it uses the full ranked gene list (all genes ordered by log2FC or padj) and tests whether predefined gene sets are enriched at the top or bottom. This could reveal coordinated transcriptional changes in GCR-exposed samples that don't meet the 2-fold cutoff individually.

Additionally, GSEA could be run with tissue-specific gene sets (the tissue-specific genes identified by Tau) as the gene sets of interest. This would directly test whether tissue-specific gene sets are enriched in the ranked DEG list, without needing to first call individual DEGs.

Tools: fgsea (R), clusterProfiler::GSEA. Input: the full DESeq2 results tables (all genes, not just DEGs), ranked by stat or log2FC.

Direction 3: Time-course analysis for temporal tissue dynamics

OSD498, OSD508, OSD510, and OSD782 all have multiple timepoints (10 min to 1440 min / 72 h). The current analysis collapses across time. A time-course analysis could reveal whether tissue-specific responses appear early (DNA damage signaling in Root within minutes) or late (growth changes in Seedling after hours).

Approaches:

- maSigPro or ImpulseDE2: time-series DE detection with tissue interaction
- WGCNA: co-expression network modules that correlate with time x tissue
- Split the current DEG sets by timepoint and re-run tissue enrichment per timepoint

Direction 4: Weighted Gene Co-expression Network Analysis (WGCNA)

WGCNA clusters genes into co-expression modules based on expression correlation across all samples. Each module can then be tested for association with traits (radiation type, dose, genotype, timepoint) and for enrichment of tissue-specific genes. This could reveal modules that are tissue-specific and radiation-responsive - the intersection of both properties.

Key question WGCNA can answer: Are there gene modules that are co-expressed specifically in one tissue and jointly upregulated by radiation? This is a stronger form of tissue-specificity than the single-gene Tau index.

Tools: WGCNA (R). Input: the raw count matrix (variance-stabilized transform) + factors_matrix.csv as trait data.

Direction 5: Single-cell or spatial transcriptomics validation

The AtGenExpress atlas is a bulk microarray dataset - it tells us which tissue a gene is predominantly expressed in, but not which specific cell types within that tissue are responding. Single-cell RNA-seq of irradiated Arabidopsis roots (e.g., using the 10x Genomics platform) would provide cell-type-level resolution of the radiation response.

Existing resources: The Arabidopsis root single-cell atlas (Denyer et al. 2019, Ryu et al. 2019) provides cell-type markers that could be used to deconvolve the bulk RNA-seq data and estimate which cell types within each tissue are contributing to the DEG signal.

Direction 6: Cross-species comparison with mammalian radiation responses

Ionizing radiation activates conserved DNA damage response pathways (ATM/ATR, p53/SOG1). Comparing the tissue-specific radiation response in Arabidopsis with published mammalian radiation atlases could reveal which aspects of the tissue-specific response are conserved (DNA repair) versus plant-specific (glucosinolate pathway, photosynthesis-related genes).

Resources: TGCA radiation exposure data, NASA Twin Study transcriptomics, and the NASA GeneLab rodent radiation studies (OSD-37, OSD-48, OSD-63).

Direction 7: Lower the fold-change threshold for exploratory analysis

The $|\log_2FC| \geq 1$ threshold is standard but conservative. Re-running with $|\log_2FC| \geq 0.5$ would approximately double the DEG count (based on the old vs new comparison: 6,141 of 6,155 lost DEGs had $|\log_2FC| < 1$). This would provide enough DEGs for GCR40/GCR80 tissue enrichment tests and could reveal subtler tissue-specific patterns. The tradeoff: more false positives and less biologically meaningful individual genes.

To do this: change LFC_THRESHOLD in the R script from 1.0 to 0.5 and re-run. The script is fully parameterized.

Direction 8: Formal interaction testing between tissue and radiation type

The current cross-comparison tests (CMH, chi-square) compare tissue response profiles across conditions, but they do not formally test the interaction between tissue and radiation type. A logistic regression model with an interaction term would directly test whether the effect of radiation on DEG probability depends on tissue:

```
glm(DEG ~ tissue + radiation_type + tissue:radiation_type, family = binomial)
```

A significant interaction term would formally prove that the tissue specificity of the radiation response differs by radiation type - i.e., roots respond more to Cs-137 while seedlings respond more to Co-60, beyond what would be expected from additive effects.

Direction 9: Promoter motif analysis of tissue-specific DEGs

If different tissues respond differently to radiation, the regulatory mechanisms may differ. Extracting promoter sequences (1 kb upstream) of tissue-specific DEGs and scanning for known motif enrichment could reveal tissue-specific regulatory elements. For example, SOG1-binding sites may be enriched in the promoters of Seedling DEGs (which are SOG1-dependent) but not in Root DEGs.

Tools: MEME Suite, PlantCARE, or ATHENA for Arabidopsis-specific motif analysis. Input: gene_id lists from the per-organ DEG CSVs.

Direction 10: Protein-protein interaction network analysis

Mapping tissue-specific DEGs onto the Arabidopsis protein-protein interaction network (AI-1, BAR interactome) could reveal whether DEGs in different tissues occupy different network neighborhoods. Tissue-specific DEGs that are also network hubs would be strong candidates for key regulators of the tissue-specific radiation response.

Tools: Cytoscape + STRING Arabidopsis network, or the Bioconductor BioNet package for subnetwork extraction.

4. Limitations of the Current Analysis

- 1. Bulk tissue inference:** The AtGenExpress atlas is a bulk microarray dataset. 'Tissue-specific' means a gene is predominantly expressed in one tissue at the whole-tissue level - it does not mean the gene responds in that tissue specifically. A gene could be expressed mainly in roots but respond to radiation in all tissues. The Tau index measures expression specificity, not response specificity.
- 2. Atlas is developmental, not stress-responsive:** The AtGenExpress atlas profiles normal development, not radiation stress. Tissue assignment reflects where a gene is normally expressed, not where it functionally responds to radiation. A stress-specific atlas would be more appropriate but does not exist at this coverage level for Arabidopsis.
- 3. ATH1 microarray vs RNA-seq platform mismatch:** The atlas uses the ATH1 Affymetrix microarray (22,810 probes), while the DEG data is from RNA-seq. Probe-to-gene mapping may miss genes not represented on the ATH1 array, and expression values are not directly comparable between platforms. 20,833 genes were mapped after deduplication.
- 4. Low-DEG comparisons:** GCR40 (2 DEGs) and GCR80 (11 DEGs) had insufficient DEGs for tissue enrichment statistics. Conclusions about tissue-specific responses to simulated GCR cannot be drawn from this analysis. GSEA (Direction 2) would address this.
- 5. Whole-seedling experiments:** All 6 experiments used whole seedlings, so the tissue-specific signal is inferred from the atlas, not directly measured. Experiments that physically separate tissues before sequencing would provide direct evidence.
- 6. Chi-square approximation warnings:** Some direction chi-square tests had small expected counts, triggering R's approximation warning. Fisher's exact test (used for per-tissue tests) does not have this issue, but the direction tests for smaller comparisons should be interpreted with caution.

5. Bottom Line

Answer to the Central Question

Yes, there are statistically significant tissue-specific responses to ionizing radiation in Arabidopsis. The evidence is strongest for Co-60 gamma radiation (chi-sq $p=2.2e-49$), where Seedling genes are 3.7x more likely to be differentially expressed. The response is tissue-dependent (different organs have different DEG rates), condition-dependent (the tissue profile differs across radiation types and doses), and genotype-dependent (sog1-1 reverses the direction and shifts the tissue distribution of the response). The most productive next steps would be GO/KEGG enrichment per tissue (Direction 1), GSEA for the low-DEG GCR comparisons (Direction 2), and formal interaction testing (Direction 8).

Files for Further Analysis

All outputs needed for the future directions above are already exported:

- Per-organ DEG gene lists: tissue_specific_degs_v2/<label>/<organ>/all_DEGs.csv
- Full DESeq2 results (all genes, ranked): all_degs_with_tissue_annotation.csv per comparison
- Tissue annotations with Tau scores: included in every master CSV
- Cross-comparison summary: cross_comparison_tissue_summary.csv
- R script (fully self-contained): tissue_specific_deg_analysis_v2.R